Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Physical vector spaces for quantum mechanics

Files.

- File: `briefs/QuantumVectorSpaces/QuantumVectorSpaces.lean`
- Library: `QuantumVectorSpaces`
- Imports: `SchwartzNorms` (the L^2 seminorm family on $\mathcal{S}(\mathbb{R})$) and new `ForMathlib` files:
  `ForMathlib/Analysis/InnerProductSpace/Defs.lean` (the seminorm $\sqrt{\mathrm{re}\langle x, x\rangle}$ of a `PreInnerProductSpace.Core`, and continuity of the inner product against it),
  `ForMathlib/Analysis/CStarAlgebra/Matrix.lean` (`Matrix.toEuclideanLin` is multiplicative),
  `ForMathlib/Analysis/Distribution/SchwartzSpace/Complete.lean` (`SchwartzMap.instCompleteSpace`: $\mathcal{S}(E, F)$ is complete for complete F),
  `ForMathlib/Analysis/Distribution/SchwartzSpace/SupNorm.lean` (`SchwartzMap.exists_norm_le_toLp_two`: a Sobolev bound for the values of a Schwartz function, from TauCeti's `SchwartzMap.exists_integral_norm_fourierInv_le`),
  `ForMathlib/Analysis/InnerProductSpace/MultilinearBasis.lean` (`ContinuousMultilinearMap.norm_le_sum_orthonormalBasis`: a multilinear map is bounded by its values on an orthonormal basis)

Setting. V is a complex vector space with an inner product given as a `PreInnerProductSpace.Core`. Unlike Mathlib's `InnerProductSpace`, this puts no norm or topology on V. The defining observables are a family X_i ($i \in \iota$) in `Module.End`, and `algebra` is the unital algebra they generate (`Algebra.adjoin`). The seminorms $p_A(v) = \sqrt{\langle Av, Av\rangle}$ form `seminormFamily`, and `topology` is the topology they generate. A topology is `IsAdmissible` if it makes V a topological vector space and makes the X_i and the inner product (jointly) continuous. For Example 3, $Xf = xf$ and $Pf = -i\hbar f'$ act on Mathlib's $\mathcal{S}(\mathbb{R}, \mathbb{C})$ (`posOp`, `momOp`), with the L^2 inner product `l2Core`. For Example 4, $S_i = (\hbar/2)\sigma_i$ acts on $\mathbb{C}^2$ as `EuclideanSpace` (`spin`). In n dimensions, $X^i f = x_i f$ and $P_i f = -i\hbar\,\partial_{e_i} f$ act on $\mathcal{S}(\mathbb{R}^n, \mathbb{C})$, with $\mathbb{R}^n$ as `EuclideanSpaceâDİ(Finn)` (`posOpN`, `momOpN`, indexed by `ObsNn`), and the L^2 inner product `l2CoreN`.

Definition 1 is `IsQuantumVectorSpacecXÄğC`: the X_i are symmetric, they satisfy $[X_i, X_j] = i\hbar \sum_k C_{ijk} X_k$ with finitely many nonzero C_{ijk} for each i, j, and V with `topology` is complete and second-countable. Completeness refers to `uniformSpace`, the uniform structure that every topological vector space carries.

Results.

Brief	Lean
Corollary 2, coarsest topology	`isGreatest_topology` (from `isAdmissible_topology` and `IsAdmissible.continuous_algebra`)
Corollary 2, locally convex	`locallyConvexSpace`
finite dimension (Example 4)	`topology_eq_moduleTopology`, `topology_qubit`
Example 3, $[X, P] = i\hbar$	`posOp_mul_momOp_sub`
Example 3, X, P, I symmetric	`isSymmetric_obs`
Example 3, Remark: the topology is that of $\mathcal{S}$	`topology_schwartz`, `withSeminorms_schwartz`, `isAdmissible_schwartz` (for $\hbar \neq 0$)
Example 4, $[S_i, S_j] = i\hbar\,\epsilon_{ijk} S_k$	`spin_comm`
Example 4, S_i symmetric	`isSymmetric_spin`
Example 4 is a quantum vector space	`isQuantumVectorSpace_qubit`
Example 3, second countability	`secondCountableTopology_schwartz`
Example 3, completeness	`SchwartzMap.instCompleteSpace`
Example 3 is a quantum vector space	`isQuantumVectorSpace_schwartz` (for $\hbar \neq 0$)
Example 3 in n dimensions, commutation relations	`posOpN_comm`, `momOpN_comm`, `posOpN_comm_momOpN`
Example 3 in n dimensions, symmetric observables	`isSymmetric_obsN`
Example 3 in n dimensions, Remark: the topology is that of $\mathcal{S}$	`topology_schwartzN` (for $\hbar \neq 0$)
Example 3 in n dimensions, second countability	`secondCountableTopology_schwartzN`
Example 3 in n dimensions is a quantum vector space	`isQuantumVectorSpace_schwartzN` (for $\hbar \neq 0$)
Theorem 5, [B]COBS	`IsAdmissible.continuous_expectation`

Differences from the brief.

- "Coarsest" is `IsGreatest` in Mathlib's order on topologies, where coarser is larger.

- The inner product is only assumed positive semidefinite in Corollary 2; definiteness is used only in finite dimension, to make the topology Hausdorff.

- Example 3 is formalized twice. For $\mathcal{S}(\mathbb{R})$, the proof that the topologies agree has two halves. (a) Each seminorm $\|x^a\partial^b f\|_2$ equals $|\hbar|^{-b}p_{X^aP^b}(f)$. (b) Every element of the algebra is a combination of the normally ordered monomials X^aP^b (`algebra_le_monoSpan`, from $PX^{n+1} = X^{n+1}P - (n+1)i\hbar X^n$), so every p_A is bounded by finitely many $\|x^a\partial^b f\|_2$. With `withSeminorms_l1_l2` from the previous brief, this gives the Schwartz topology.

- For $\mathcal{S}(\mathbb{R}^n)$, one half is that the Schwartz topology is admissible (`isAdmissible_schwartzN`), hence finer than the topology of Corollary 2. For the other half, each Schwartz seminorm $\sup_x \|x\|^k\|D^m f(x)\|$ is bounded by finitely many p_A (`exists_seminorm_le`). The derivative $D^m f(x)$ is bounded by its values on the unit vectors, $\|x\|^k \le (1+\|x\|^2)^k$ is the polynomial $(1+\sum_i (X^i)^2)^k$, and the Sobolev bound $|g(x)| \le C(\|g\|_2 + \sum_i \|\partial_i^{n+1} g\|_2)$ turns the supremum into L^2 norms of elements of the algebra applied to f. No normal ordering is needed, since the operators $\partial_i^{n+1}(1+\sum_j (X^j)^2)^k\partial^r$ are themselves in the algebra.

- In finite dimension the topology of Corollary 2 is the unique Hausdorff vector-space topology, whatever the defining observables (`topology_eq_moduleTopology`). This is how Example 4 is checked.

- Second countability in n dimensions comes from a general fact (`secondCountableTopology_topology`). With countably many defining observables, the p_A for the words $A = X_{i_1}\cdots X_{i_k}$ already generate the topology, so $v \mapsto (Av)_A$ embeds V into a countable product of copies of L^2.

- Second countability of $\mathcal{S}(\mathbb{R})$ is proved by embedding it into a countable product of copies of $L^2(\mathbb{R})$ through $f \mapsto (x^a\partial^b f)_{a,b}$ (`l2Embedding`). The topology of $\mathcal{S}(\mathbb{R})$ is induced by this map, because it is given by the L^2 seminorms (`withSeminorms_l1_l2`).

- Completeness of $\mathcal{S}(E,F)$, for any complete F, is proved in `ForMathlib`, since neither Mathlib (at this pin or on master) nor TauCeti nor physlib has it. The proof is the classical one (Becnel and Sengupta, *Mathematics* 3 (2015), §3.5): along a Cauchy sequence the weighted derivatives $\|x\|^k D^n u_m$ converge uniformly; the limits are the derivatives of the limit function, by the theorem on uniform limits of derivatives (`hasFDerivAt_of_tendstoUniformly`); and the bounds pass to the limit. For the qubit, completeness and second countability come from finite dimension.

- Not formalized: the "iff $n = m$" part of Example 3, which needs a notion of isomorphism of quantum vector spaces; [B]WDQ and [B]STDIST in Theorem 5 and the remarks of section 5, which are informal.

Problems found in the brief. No errors. Two steps are left implicit.

- The Remark after Example 3 says the L^2 seminorms and the sup seminorms dominate each other. But the seminorms of Corollary 2 are $\|Af\|$ for every A in the algebra, including non-normally-ordered products such as PX. Reducing them to $\|x^a\partial^b f\|$ needs the commutation relation (normal ordering), which the brief does not mention.

- Example 3 needs $\hbar \neq 0$: for $\hbar = 0$ we get $P = 0$, and the topology is just that of L^2.